Customer Shopping Behavior Analysis

Data preparation in Python, business analysis in SQL, and an interactive Power BI dashboard

1. Project Overview

This project analyzes the shopping behavior of a retail company's customers to answer one business question: how can the company use consumer data to identify trends, improve customer engagement, and optimize its marketing and product strategies? The work covers data preparation in Python, ten business queries in PostgreSQL, an interactive Power BI dashboard, and this report with its recommendations.

2. Dataset Summary

- Source: customer_shopping_behavior.csv
- Rows: 3,900 purchases, each with a unique customer ID
- Columns: 18 in the raw file, 20 after cleaning (two added, one dropped, one renamed)
- Customer demographics: age, gender, location (50 US states), subscription status
- Purchase details: item purchased (25 items), category (4), purchase amount in USD, size, color, season
- Shopping behavior: discount applied, promo code used, previous purchases, purchase frequency, review rating, shipping type, payment method
- Missing data: 37 null values in Review Rating, none in any other column
- Duplicates: none

Overall figures: total revenue is $233,081, the average purchase is $59.76, and the average review rating is 3.75.

3. Data Preparation in Python

Cleaning and transformation were done in a Jupyter notebook using pandas.

- **Loading and inspection.** The CSV was loaded and checked with head(), columns and isnull().sum().
- **Missing values.** The 37 missing review ratings were filled with the median rating of the product category each purchase belonged to. A category median was used instead of a global mean so that categories with different rating levels were not pulled toward each other.
- **Column standardization.** All column names were converted to lowercase snake_case, and purchase_amount_(usd) was renamed to purchase_amount.
- **Age groups.** A new age_group column was created with pd.qcut into four equal-sized bins: Young Adult (18 to 31), Adult (32 to 44), Middle-aged (45 to 57) and Senior (58 to 70).

- **Purchase frequency.** A new purchase_frequency_days column converts the text frequency (Weekly, Fortnightly, Monthly, Quarterly, Annually and so on) into a number of days between purchases.
- **Redundant column.** discount_applied and promo_code_used matched in every row, so promo_code_used was dropped.
- **Duplicate column fix.** Re-running the imputation step created a second review_rating column, which was removed before loading.
- **Database load.** The cleaned DataFrame was written to a PostgreSQL table named customer using SQLAlchemy and psycopg2.

4. Data Analysis in SQL

Ten business questions were answered in PostgreSQL against the customer table.

Q1. Revenue by gender

Gender	Revenue
Female	$75,191
Male	$157,890

Males make up 68% of the dataset (2,652 of 3,900), so the revenue gap mostly reflects customer count. Average spend per purchase is nearly equal: $59.54 for males and $60.25 for females.

Q2. Discount users who spent above average

839 purchases used a discount and still exceeded the overall average of $59.76. That is about half of the 1,677 discounted purchases.

Q3. Top 5 products by average rating

Product	Average rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

The spread across the top five is only 0.08, so the ranking is tight.

Q4. Standard vs. Express shipping

Shipping type	Average purchase
Standard	$58.46
Express	$60.48

Q5. Subscribers vs. non-subscribers

Subscribed	Customers	Avg spend	Total revenue
Yes	1,053	$59.49	$62,645
No	2,847	$59.87	$170,436

Subscribers do not spend more per purchase. Non-subscribers bring in more revenue only because there are more of them.

Q6. Products with the highest discount share

Product	Purchases with discount
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Q7. Customer segments by previous purchases

New is 1 previous purchase, Returning is 2 to 10, and Loyal is more than 10.

Segment	Customers
Loyal	3,116
Returning	701
New	83

Q8. Top products within each category

Category	1st	2nd	3rd
Accessories	Jewelry (171)	Sunglasses (161)	Belt (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	only two items

Q9. Repeat buyers and subscription

Among customers with more than 5 previous purchases, 2,518 are not subscribed and 958 are. As a share, 27.6% of repeat buyers subscribe, against 22.4% of customers with 5 or fewer previous purchases. The relationship is weak.

Q10. Revenue by age group

Age group	Age range	Revenue
Young Adult	18 to 31	$62,143
Middle-aged	45 to 57	$59,197
Adult	32 to 44	$55,978
Senior	58 to 70	$55,763

5. Dashboard in Power BI

An interactive dashboard titled Customer Insights Dashboard was built in Power BI on the cleaned data and presents the results on a single page.

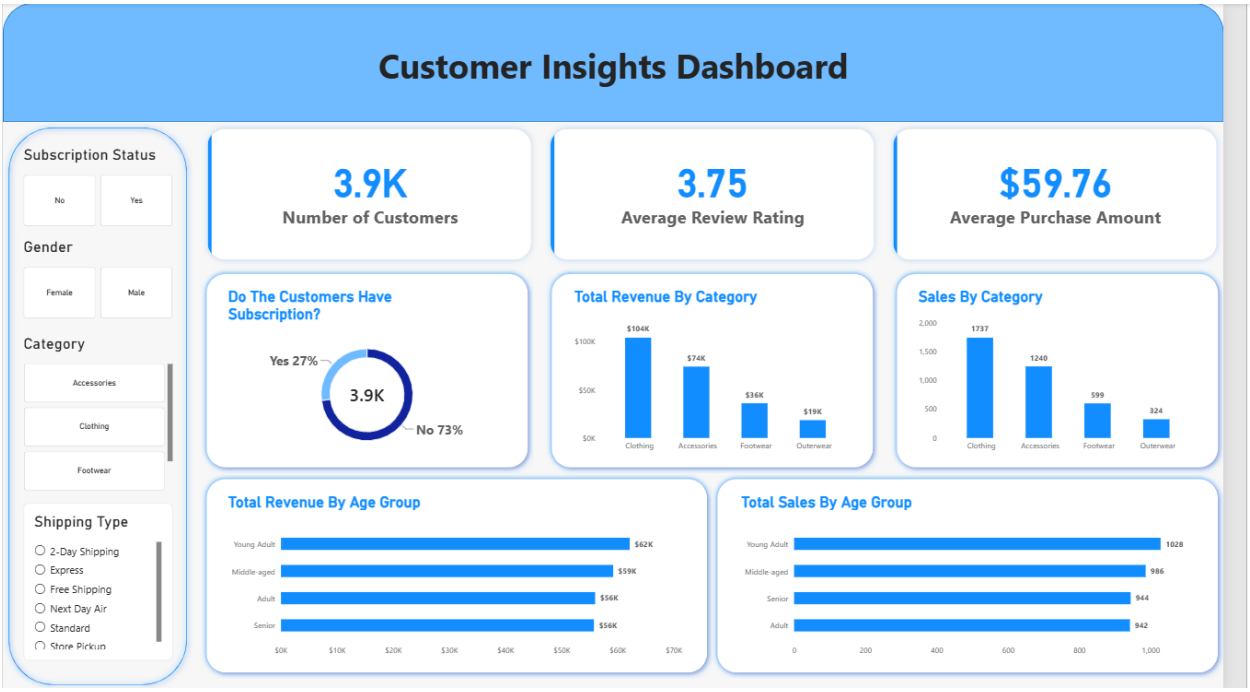


Figure 1. Customer Insights Dashboard

- **Summary cards.** 3.9K customers, an average review rating of 3.75, and an average purchase amount of $59.76.
- **Subscription split.** 27% of customers are subscribed and 73% are not.
- **Category view.** Clothing is the largest category on both measures, with $104K revenue from 1,737 sales. Accessories follow with $74K and 1,240 sales, then Footwear with $36K and 599 sales, and Outerwear with $19K and 324 sales.
- **Age group view.** Young Adult is the top group with $62K revenue and 1,028 sales, followed by Middle-aged with $59K and 986 sales. Adult and Senior are close at about $56K each, with 942 and 944 sales.

- **Filters.** Slicers for subscription status, gender, category and shipping type update every chart on the page, so a stakeholder can, for example, look at only female customers buying Footwear with Express shipping.

6. Key Findings

- **Spend is flat across segments.** Average purchase value sits between $57 and $61 for every gender, category, age group, subscription status and shipping type.
- **Revenue differences come from volume.** Clothing leads with $104,264 across 1,737 purchases, and Outerwear is the smallest with $18,524 across 324. Male customers lead for the same reason: there are more of them.
- **Subscribers are a minority and not more valuable per purchase.** Only 27% of customers subscribe, and their average spend is slightly below non-subscribers.
- **Discounts are used widely.** 43% of purchases (1,677 of 3,900) had a discount. Discounted purchases average $59.28 against $60.13 without one, so the discount does not appear to raise basket size.
- **Young adults are the top revenue group,** but the gap to the lowest group is about $6,400, roughly 11%.
- **Seasonal revenue is even.** Fall is highest at $60,018 and Summer lowest at $55,777.
- **Subscription by gender.** Every female customer in the dataset has subscription status No, while 40% of male customers are subscribed.

7. Business Recommendations

1. **Rework the subscription offer.** Subscribers spend no more than other customers, so the current benefits are not changing behavior. Introduce benefits tied to spend or frequency, then track whether subscriber spend moves.
2. **Build loyalty around real differences.** With 80% of customers in the Loyal segment, the current threshold separates almost no one. Use finer tiers, for example 11 to 25, 26 to 40 and 41 or more previous purchases.
3. **Review the discount policy.** Discounts are applied to nearly half of all purchases without a lift in purchase value. Test reducing them on the most discounted items (Hat, Sneakers, Coat, Sweater, Pants) and compare margin.
4. **Feature top-rated and top-selling products.** Use Gloves, Sandals, Boots, Jewelry and Blouse in campaigns, since they combine strong ratings or high order counts.
5. **Target marketing by volume.** Young Adult and Middle-aged customers bring the most revenue, and Clothing and Accessories drive most sales. Concentrate campaigns there.
6. **Look at female customers and subscriptions.** Female customers are 32% of the base and none of them subscribe. This needs a separate check on how the subscription is offered.

8. Limitations

- The dataset has no dates, so trends over time and repeat-purchase timing cannot be measured. Seasonality only reflects the season of each purchase.
- Each row is one purchase with its own customer ID, so customer lifetime value cannot be calculated.
- Age groups are quartiles of this dataset, not standard age brackets.
- Differences between groups are small and no statistical tests were run, so some gaps may not be significant.

9. Tools Used

Python (pandas, SQLAlchemy, psycopg2), PostgreSQL, Power BI, Jupyter Notebook, Gamma, GitHub.